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MMP-8 Is the Predominant Collagenase in Healing Wounds and Nonhealing Ulcers[1]

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Background. **The initial cleavage of collagen by collagenase represents the rate-limiting step in the degradation of this central extracellular matrix protein. Chronic nonhealing ulcers, especially pressure ulcers, typically contain elevated levels of collagenolytic activity. However, there have been no detailed attempts to identify the source of these collagenases and their activity either in normal healing wounds or in chronic nonhealing ulcers.**

Materials and Methods. **Levels of the matrix metalloproteinases, MMP-1 and MMP-8, and the tissue inhibitor of matrix metalloproteinases, TIMP-1, were measured in fluids and tissues of healing human wounds and nonhealing ulcers by ELISA. Relative MMP-1 and MMP-8 levels were also analyzed by substrate preference in a functional assay.**

Results. **The patterns of the collagenases MMP-1 and MMP-8 in healing wounds were distinct, with MMP-8 appearing in significantly greater amounts than MMP-1. Chronic nonhealing ulcers were characterized by significantly higher levels of MMP-1 and MMP-8, and lower levels of TIMP-1, than in healing wounds. Levels of both MMP-1 and MMP-8 varied greatly in chronic ulcers, although MMP-8 was always the predominant collagenase present in these wounds. Interestingly, these collagenases were present almost exclusively in their inactive forms in healing wounds, whereas nonhealing ulcers possessed significant levels of the active forms of these enzymes.**

Conclusions. **These results clearly demonstrate that the neutrophil-derived MMP-8 is the predominant collagenase present in normal healing wounds and suggest that overexpression and activation of this collagenase may be involved in the pathogenesis of nonhealing chronic ulcers. In addition, excessive collagenolytic activity in chronic ulcers is made possible, partly because of the reduced levels of the inhibitor, TIMP-1.** **© 1999 Academic Press**

Key Words: **wound healing; chronic wounds; neutrophil; matrix metalloproteinases.**

INTRODUCTION

Matrix degradation is an essential component of the normal wound repair process and is involved in wound debridement, dissolution of the basement membrane, ingrowth of capillary buds, turnover of the extracellular matrix, and subsequent long-term tissue remodeling. Because many chronic ulcers are characterized by massive tissue loss, it has been speculated that an imbalance in the relative activities of matrix-degrading enzymes and their inhibitors is involved in the establishment and persistence of many chronic ulcers [1–5].

Chronic ulcers, particularly pressure ulcers, have been shown to contain elevated levels of matrix-degrading enzymes including cathepsin G, plasminogen activators, elastase, and several matrix metalloproteinases (MMPs) [5–9]. Collectively, the MMPs have the ability to degrade virtually every component of the extracellular matrix, including proteoglycans. The presence of elevated levels of several MMPs in chronic wound fluids has been documented extensively [5, 6, 10, 11].

In a previous study, we showed that pressure and venous ulcers contain higher levels of collagenase compared to acute healing wounds and, perhaps more importantly, significantly elevated levels of collagenolytic activity [5]. The profound degradation of type 1 collagen substrate suggested that substantial amounts of neutrophil collagenase MMP-8 are present in chronic

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ulcers. The purpose of this study was to identify and analyze the expression of the collagenases and their inhibitor, TIMP-1, in healing normal wounds and nonhealing ulcers.

MATERIALS AND METHODS

Collection of wound fluids. Informed consent was obtained for all procedures and approval was received from the Virginia Commonwealth University Committee on the Conduct of Human Research in accordance with the 1975 Declaration of Helsinki. Healthy volunteers (11 females, 9 males; age range 19–54 years; 17 whites, 3 blacks) were recruited from the local population and the exclusion criteria included smoking, diabetes mellitus, systemic disease, use of medications, which may impair wound healing, or history of abnormal wound healing. Under a 1% lidocaine local anesthesia and with full sterile precautions, four full-thickness skin biopsies were taken from over the sacral or trochanteric regions using a 6-mm trephine (day 0). Wounds were covered with a transparent polyurethane occlusive film dressing (Tegaderm; 3M, St. Paul, MN) under which exudates accumulated. Wound exudates were aspirated through the film dressing using a 30-gauge hypodermic needle mounted on a 1-cc tuberculin syringe. Wound exudate was collected on days 1, 2, 3, 4, 7, and 8. The four wounds were subsequently biopsied, one each on days 2, 4, 14, and 28 using a 4-mm trephine placed at the wound edge. Once a wound had been thus biopsied, it was excluded from further wound exudate collection.

Drainage fluids from surgical wounds were obtained from inpatient populations at the Medical College of Virginia Hospitals who had undergone either mastectomy or myocutaneous flap reconstruction procedures ($n = 4$). Fluids were collected from Jackson–Pratt drains over a 4-h period on postoperative days 1–4.

Chronic ulcer fluids were collected from 13 patients (3 venous ulcers and 10 pressure ulcers) whose ulcers had been present for greater than 3 months and that showed healing at ≤1 mm/week. Only ulcers without clinical evidence of infection were used in this study. The ages of patients with chronic ulcers ranged from 40 to 66 years, and all had a normal nutritional status. Ulcer fluid collection was carried out as described above, over a period of 4–8 h. All samples were immediately clarified at 14,000*g* for 15 min at 4°C and the supernatant stored at −20°C until analyzed.

Determination of protein concentration. All samples were assayed for protein content using the Bradford protein assay method [12]. Bio-Rad reagent was used (Bio-Rad Laboratories, Hercules, CA) with bovine serum albumin as a standard. Subsequent analysis of all wound and ulcers fluids was normalized in regard to the protein concentration.

MMP-1, MMP-8, and TIMP-1 ELISA. Total human MMP-1 and TIMP-1 in samples was determined using the Biotrak sandwich ELISA system (Amersham, Little Chalfont, Buckinghamshire, UK). Standards and samples (100 μg protein) were incubated in microtiter wells precoated with the respective mouse primary antibody followed by a secondary rabbit polyclonal antibody. The resulting antigen–antibody complex was detected using donkey anti-rabbit horseradish peroxidase and the conjugate was quantified by a colorimetric reaction with 3,3′,5,5′-tetramethylbenzidine substrate and the resultant color was read at 450 nm (Softmax spectrophotometer, Molecular Devices, Palo Alto, CA). All samples were assayed in duplicate and the concentration of target protein in each sample was determined by interpolation from the standard curve. Similarly, levels of MMP-8 were measured using an ELISA kit (Fuji Chemicals, Toyama, Japan). Differences in levels of MMPs between surgical drainage, healing dermal wounds, and chronic nonhealing ulcers were tested for statistical significance using a one-way analysis of variance (ANOVA), whereas those between MMP-1 and MMP-8 were analyzed using a two-tailed Student's *t* test. These ELISAs detect both the active and inactive forms of MMP-1 or MMP-8 and their lower level of detection was 0.6 ng of protein. They do not detect either enzyme when complexed with α_2-macroglobulin or TIMP.

Specificity of the MMP-1 and MMP-8 ELISAs was assessed by determining their relative abilities to detect collagenases produced by purified neutrophils (a source of MMP-8) or by appropriately stimulated dermal fibroblasts (a source of MMP-1). Human dermal fibroblasts were stimulated with a combination of IL-1β and PDGF-BB to induce the expression and secretion of MMP-1. Freshly isolated neutrophils were incubated in medium containing 100 nM phorbol ester to induce release of MMP-8 from secondary granules. ELISA did not detect the presence of MMP-1 in neutrophil-conditioned medium even at concentrations 100-fold greater than that used in the MMP-8 ELISA. Similarly, the MMP-8 ELISA did not respond to fibroblast conditioned medium.

Collagenase activity assays. Collagenolytic assays using types I and III collagen substrates were performed as described previously [5]. Briefly, fetal (day 26 gestation) rabbit skin was incubated in Dulbecco's modified Eagle's medium (DME) containing 10 μCi/ml each of ^{14}C-labeled proline [246 mCi/mmol] and glycine [44 mCi/mmol] (Moravek Biochemicals, Inc., Brea, CA). Collagen was extracted by pepsin (Sigma Chemical Co., St. Louis, MO) digestion in 0.5 N glacial acetic acid for 48 h at 4°C. ^{14}C-labeled type III and I collagens were selectively precipitated by dialysis against 50 mM Tris–Cl containing 1.8 and 3.5 M NaCl, respectively. Samples were analyzed in triplicates with 50 μg protein used in each reaction. Samples were incubated with the radiolabeled collagen substrates at 30°C for 1, 4, and 24 h each. Negative controls contained 10 mM ethylenediaminetetraacetic acid (EDTA), to inhibit all collagenolytic activity. To determine total collagenase activity, 1.0 mM *p*-aminophenyl mercuric acetate (APMA) (Sigma Chemical Co.), an activator of the enzyme, was included in parallel reactions. Reactions were terminated by adding Laemmli's loading buffer and the products were examined by electrophoresis on 6% sodium dodecyl sulfate–polyacrylamide gels. Protein bands on the gel were detected by fluorography and archived using a GDS75000 gel documentation system (UVP, Upland, CA).

Conditioned medium from phorbol ester-stimulated human dermal fibroblasts or neutrophils was used as a source of MMP-1 and MMP-8. Fibroblasts were established by explanting dermis from skin, which had been removed for cosmetic purposes. Fibroblast lines were maintained in DME supplemented with penicillin (100 units/ml), streptomycin (100 μg/ml), and 10% fetal bovine serum. Lines were routinely examined for the presence of mycoplasma (MYCOTECT Kit, Life Technologies, Gaithersburg, MD) at the fourth passage. Confluent cultures were exposed overnight to a combination of 10 ng/ml interleukin-1β (IL-1β) and 50 ng/ml platelet-derived growth factor-BB (PDGF-BB) (Genzyme Diagnostics, Cambridge, MA) in DME supplemented with antibiotics and 1% ITS$^+$ (Collaborative Biomedical Products, Bedford, MA). The conditioned medium was concentrated 10-fold using Centriprep-10 concentrators (Amicon, Inc., Beverly, MA). Neutrophils were isolated from blood obtained from volunteers. Blood was collected by venipuncture and neutrophils were isolated in a single step by density gradient centrifugation through a cushion of sucrose polymer and sodium diatrizoate (NIM) using the manufacturer's recommended protocol (Cardinal Associates, Inc., Sante Fe, NM). To induce degranulation, isolated neutrophils were incubated 30 min at 37°C in DME containing 100 nM 12-*O*-tetradoecanoylphorbol 13-acetate followed by removal of the cells by centrifugation (1000*g* for 5 min at 4°C).

RESULTS

MMP-8 Is the Predominant Collagenase in Healing Normal Wounds and Nonhealing Ulcers

Results of ELISA analyses of MMP-1 and MMP-8 levels in exudates from healing dermal wounds, drain-

TABLE 1

Collagenase Levels in Wound Fluids Obtained from Surgical Wounds, Healing Open Human Dermal Wounds, and Nonhealing Chronic Ulcers[a]

	MMP-1	MMP-8
Surgical wound fluid (n = 4)		
Day 1	1.2 (1.1)	420 (415)
Day 2	<0.6	275 (234)
Day 3	<0.6	66 (40)
Dermal wound fluid (n = 20)		
Day 1	0.7 (1.1)	2130 (1180)
Day 2	<0.6	1680 (639)
Day 3	<0.6	2530 (990)
Day 4	2.7 (2.3)	4340 (1600)
Day 7	15.9 (11.6)	3330 (918)
Day 8	6.8 (9.8)	1860 (870)
Chronic ulcer fluid (n = 13)	45 (23)	6010 (680)

[a] Matrix metalloproteinase (MMP-1 and MMP-8) levels measured by ELISA are expressed as ng/100 μg protein. Each data point represents the mean collagenase level (±SD) for each group of subjects.

age fluids from surgical wounds, and chronic nonhealing wounds are presented in Table 1 and in Figs. 1 and 2. At each corresponding time point, levels of either collagenase were significantly lower in surgical drainage fluid than in fluids collected from healing dermal wounds. In addition, mean levels of either MMP-1 or MMP-8 protein were significantly higher in fluids from chronic nonhealing ulcers than at any time in healing dermal wounds. Regardless of the wound type, levels of MMP-8 were at least 100-fold higher than MMP-1, indicating that MMP-8 is the predominant collagenase present in both normal and pathologic conditions.

Functional Assays Indicate Increased Collagenolytic Activity in Chronic Wounds

To corroborate the results obtained by ELISA, advantage was taken of differences in the substrate preferences between MMP-1 and MMP-8. It is well established that MMP-8 (neutrophil collagenase) degrades type I collagen in preference to type III collagen, whereas MMP-1 (fibroblast-derived collagenase) is more active against type III collagen than against type I [13–18]. An example of this substrate preference is shown in Fig. 3. Functional assays using types I and III collagen as substrates were performed with neutrophil- and fibroblast-conditioned medium. Following activation with APMA, fibroblast-conditioned medium degraded type III collagen to a greater extent than type I collagen, whereas neutrophil-conditioned medium preferentially cleaved type I collagen.

Collagenase activity assays were carried out with fluid samples from surgical drainage, normal healing dermal and chronic nonhealing ulcers using types I

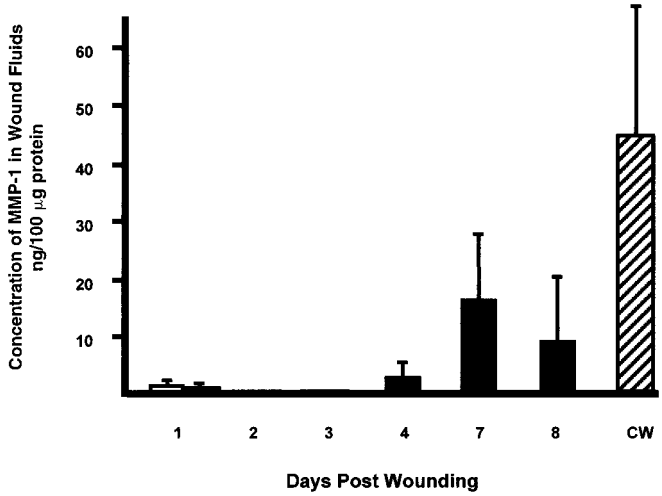


FIG. 1. Quantitative analysis of MMP-1 protein levels in fluids obtained from surgical wounds, healing open dermal wounds, and nonhealing pressure ulcers. Levels of MMP-1 were determined in 100-μg aliquots of wound fluids using ELISA analysis. Surgical wound fluids (n = 4) collected on days 1–3 postsurgery are represented as open bars. Dermal wound fluids (n = 20) collected on days 1–4, 7, and 8 are represented as black bars. Mean levels of MMP-1 in pressure ulcer fluids (n = 23) are represented as a cross-hatched bar. Error bars represent standard deviation.

and III collagen as substrates. As shown in Fig. 4, degradation of type I collagen occurred after only 1 h incubation with fluids from healing dermal wounds. However, the characteristic TC^A products were generated in significant amounts only with APMA activa-

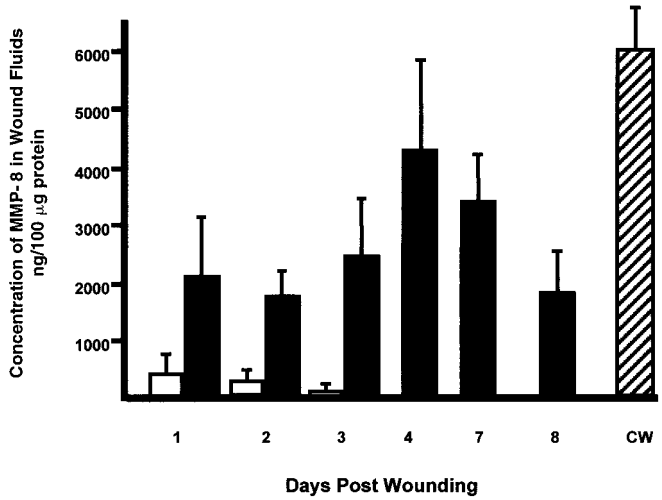


FIG. 2. Quantitative analysis of MMP-8 protein levels in fluids obtained from surgical wounds, healing open dermal wounds, and nonhealing pressure ulcers. Levels of MMP-8 were determined in 100-μg aliquots of wound fluids using ELISA analysis. Surgical wound fluids (n = 4) collected on days 1–3 postsurgery are represented as open bars. Dermal wound fluids (n = 20) collected on days 1–4, 7, and 8 are represented as black bars. Mean levels of MMP-8 in pressure ulcer fluids (n = 23) are represented as a cross-hatched bar. Error bars represent standard deviation.

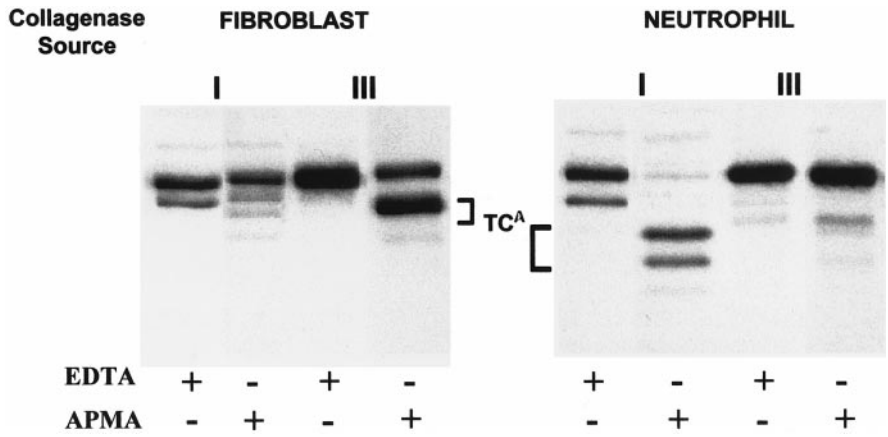


FIG. 3. Functional assay of collagenolytic activity using type I and II collagen as substrates. This assay demonstrates the substrate preferences for neutrophil collagenase (MMP-8) and fibroblast-derived interstitial collagenase (MMP-1). Following activation with APMA, fibroblast-conditioned medium degraded type III collagen to a greater extent than type I collagen while neutrophil-conditioned medium preferentially lysed type I collagen, producing typical TC^A products. Collagenase activity was uniformly abolished by the addition of EDTA.

tion. This indicates that the majority of the collagenase (MMP-8) present in exudates from healing open human wounds was in the inactive, proenzyme form. In contrast, under the same assay conditions, degradation of type III collagen was not appreciable even with APMA activation. Similar observations were made with drainage fluids obtained from surgical wounds; type I collagen degradation was noted only with APMA activation (Fig. 5), but no type III collagen degradation occurred (results not shown).

In contrast, chronic nonhealing ulcer fluids exhibited a greater degree of collagenolysis compared to fluids obtained either from healing dermal wounds or surgical wounds (Figs. 4 and 5). Degradation of type I collagen by chronic ulcer fluids occurred within 1-h incubation and the addition of APMA did not yield greater amounts of TC^A products. This indicates that collagenase in fluids from chronic ulcers was predominantly in the activated form. Again, as observed with the acute wound fluids, little degradation of type III collagen was noted with chronic ulcer fluid, even following an extended period of incubation (48 h). These data are consistent with the presence of large amounts of MMP-8 compared to MMP-1 in both acute healing and nonhealing ulcer fluids.

TIMP-1 Levels Are Diminished in Nonhealing Wounds

Levels of the MMP-specific inhibitor, TIMP-1, in healing and nonhealing ulcers were also examined. The TIMP-1 ELISA recognizes total human TIMP-1

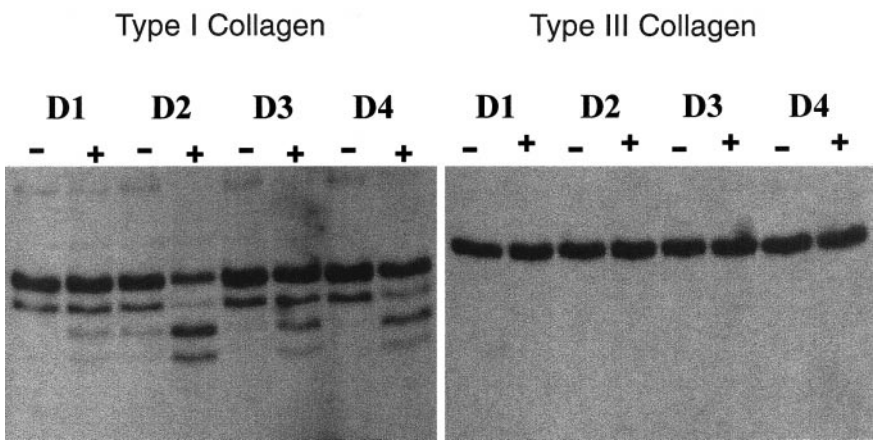


FIG. 4. Collagenase activity assay with fluid samples from open dermal wounds. Wound fluids containing 100 μg of protein were incubated with ^{14}C-labeled type I or type III collagen for 1 h at 32°C. Degradation of type I collagen occurred after only 1-h incubation in samples taken from postbiopsy day 1 (D1) through day 4 D4). However, the characteristic TC^A products were generated in significant amounts only with APMA activation (+). In contrast, under the same assay conditions, degradation of type III collagen was not appreciable even with APMA activation. These results indicate that MMP-8 is the predominate collagenase present in the fluids of healing open dermal wounds but that the majority of this enzyme exists as inactive zymogen.

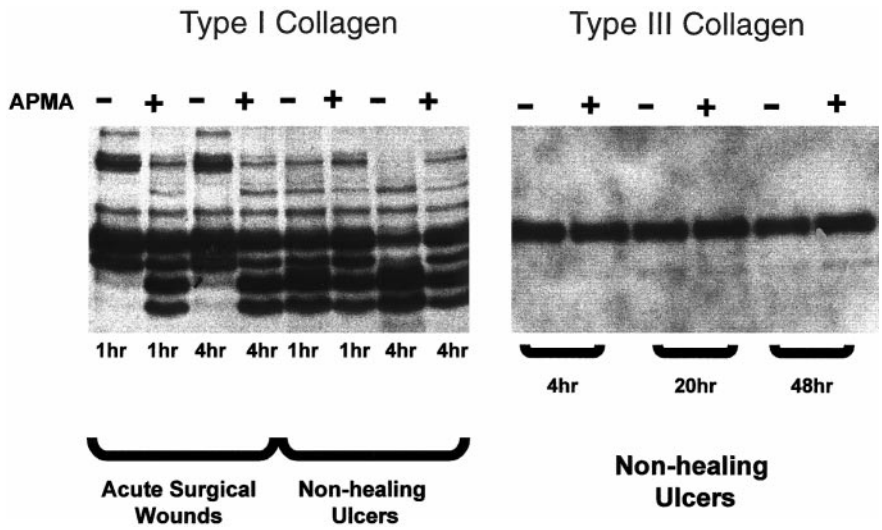


FIG. 5. Collagenase activity in fluid samples from acute surgical wounds and nonhealing chronic ulcers type I and III collagen as substrates. Fluids from healing surgical wounds or nonhealing pressure ulcers containing 100 μg of protein were incubated with ^{14}C-labeled type I or type III collagen for different times at 32°C. Without first activating with APMA, acute surgical wound fluid degraded little type I or III collagen even with prolonged incubation (4 h). However, in the presence of the MMP-activator APMA, collagen degradation occurred as indicated by the presence of the TC^{A} products. Under similar assay conditions, degradation of type III collagen was not observed. In contrast, fluids from pressure ulcers efficiently degraded type I collagen in the presence or absence of APMA.

(including free TIMP-1 and that complexed with MMPs). Using this assay, levels of TIMP-1 were found to be markedly elevated on day 1 postwounding and subsequently gradually declined with time to plasma levels (Fig. 6). In contrast, levels of TIMP-1 in ulcer fluids were either very low or not detected.

DISCUSSION

A salient feature of this study is the use of two independent assays to demonstrate that the predominant collagenase present in the fluids of both healing wounds and in chronic pressure ulcers is MMP-8. In addition, whereas active MMP-8 was readily detected in the fluids from pressure ulcers, it was more difficult to detect the active form of this enzyme in either open dermal wounds or surgical wounds. Several studies have implicated MMPs in the excessive matrix degradation associated with chronic ulcers and a variety of other pathological processes, including periodontal disease, arthritis, dilated cardiomyopathy, and cancer metastases [5, 19–23]. The majority of these studies have focused on the gelatinases and stromelysins. Elevated levels of the gelatinases MMP-2 and MMP-9 have been demonstrated in venous and pressure ulcers by means of quantitative zymography [1, 5, 10, 24–26]. Immunolocalization and *in situ* hybridization studies of healing dermal and chronic ulcers have defined the pattern of temporal and spatial expression of several MMPs, including stromelysin-1, stromelysin-2, MMP-1, and MMP-2. These studies indicate that distinct populations of keratinocytes and stromal cells express different MMPs in human wounds [27–29]. Stromelysin-1 (MMP-3) transcripts were shown to be expressed more consistently, and in larger numbers of basal keratinocytes, in chronic ulcers compared to nor-

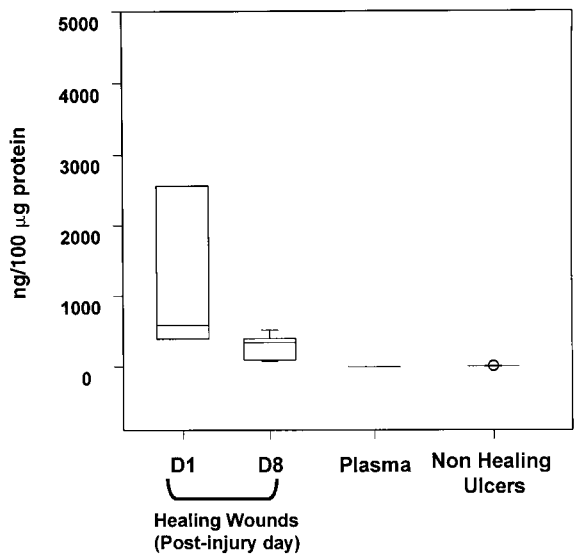


FIG. 6. Levels of the tissue inhibitor of matrix metalloproteinases-1 (TIMP-1) in plasma and fluid samples obtained from healing open human dermal wounds and nonhealing chronic ulcers. TIMP-1, as measured by ELISA, was barely detected in plasma but levels in healing wounds were greatly elevated on postinjury day 1 and diminished over time as healing proceeded. However, the levels in healing wounds (n = 20) were always greater than those in nonhealing wounds (n = 13). TIMP-1 levels at day 8 postinjury in healing wounds remained significantly greater than those in nonhealing chronic ulcers (P = 0.002, Student's t test).

mally healing wounds [11]. Localization studies have not revealed clear differences between the healing wounds and chronic ulcers, particularly for the collagenase MMP-1. In the current study, we have shown that levels of the collagenases MMP-1 and MMP-8 are significantly elevated in chronic wounds compared to healing dermal ulcers.

In this study, mean levels of MMP-8 protein in chronic ulcers fluids were found to be significantly higher than those in acute wound fluids. This finding was corroborated by two independent experimental approaches. Perhaps the finding of greatest importance is that a greater proportion of the collagenases in chronic ulcer fluids was found to be in an active state. Although MMP-1 levels were significantly greater in chronic ulcers than in the acute wounds, the amounts present were insufficient to produce appreciable degradation of type III collagen. These results suggest that the excessive collagenolytic properties of chronic ulcers fluids were predominantly due to the presence of activated MMP-8. In a previous study, mean levels of collagenolytic activity in fluids from pressure ulcers were found to be more than 20-fold higher than in fluids from surgical wounds [5]. High levels of active collagenase would have obvious implications for the ability of these ulcers to heal.

Although the expression of MMP-8 has been reported to occur in at least one other cell type, neutrophils serve as the principal source of this enzyme [13, 35]. Neutrophils are associated with the initial inflammatory stages of wound repair and this was reflected by the presence of MMP-8 in the fluids of healing wounds at early times. The inflammatory stage is extended in the open wound model and this is reflected by the continued presence of neutrophils and MMP-8 as late as 8 days postwounding. Massive loss of tissue is a hallmark of pressure ulcers. We and others have noted the extensive presence of neutrophils in pressure ulcers [36]. It is therefore not surprising that these wounds contain elevated levels of MMP-8 [5]. The gelatinase, MMP-9, is also stored in abundant amounts in the granules of neutrophils. We have previously demonstrated that levels of MMP-9 are significantly higher in chronic nonhealing pressure and venous stasis ulcers than in surgical wounds [5]. More recently, we have also found that myeloperoxidase and neutrophil elastase levels correlate well with the MMP-8 profiles that we found in these different types of wounds (unpublished observations).

Expression of transcripts for another collagenase, MMP-13, has recently been demonstrated in venous ulcers [31]. This same group of investigators was not able to detect the expression of MMP-13 in normal human tissue or healing wounds [31]. MMP-13 possesses the ability to degrade types I and III collagen to a similar (or greater) degree than MMP-1 and MMP-8, respectively. Because little type III collagen is degraded by fluids from either healing wounds or pressure ulcers, this would seem to indicate that MMP-13 is not present in abundance.

One question which is raised by this study is what is the underlying mechanism for the relatively high levels of active MMP-8 in pressure ulcers? One possible mechanism that could lead to high levels of active collagenase is an increase in the agent or agents responsible for activating the MMP-8 zymogen. At present, little is known regarding the *in vivo* activation MMP-8. Diminished levels of endogenous collagenase inhibitors might also contribute to the elevated levels of active collagenase. Intact forms of the nonspecific proteinase inhibitor, α_2-macroglobulin, and the serine proteinase inhibitor, α_1-antiproteinase, have been shown to be present in reduced levels in chronic wounds [32–34]. This study confirms previous findings of significantly reduced levels of the tissue inhibitor of metalloproteinase-1 (TIMP-1) in fluids from chronic wounds compared to that of healing surgical wounds [1, 5].

This study further highlights the limitations imposed by available wound models on the studies of enzyme profiles of acute wounds and chronic ulcers. We have shown that significant differences can exist between fluids obtained from open healing dermal wounds compared with drainage fluids from surgical wounds, which have been used in many previous studies. Regardless of the differences, it is clear that chronic ulcers exhibit significantly greater collaenolytic activity. The significance of this finding in understanding the pathogenesis of chronic ulcers remains unclear. The presence of abundant amounts of MMP-8 may reflect merely the high degree of neutrophil infiltration characteristic of most chronic ulcers. However, the striking imbalance between active collagenases and their inhibitors in these ulcers would be expected to prevent the accumulation of a collagen-rich matrix during the process of tissue repair.

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